

Policy Function Iteration

Yaolang Zhong

University of Tokyo

yaolang.zhong@e.u-tokyo.ac.jp

Recap: Markov Decision Process (MDP)

- ▶ An MDP is defined by the tuple

$$(\mathcal{S}, \mathcal{A}, P, r, \beta)$$

where:

- ▶ **State space $\mathcal{S}$** : Possible system states.
- ▶ **Action space $\mathcal{A}(s)$** : Feasible actions when in state s .
- ▶ **Transition kernel $P_t(s' | s, a)$** : Probability of moving from s to s' given action a .
- ▶ **Reward function $r_t(s, a)$** : Instantaneous payoff from taking action a in state s .
(Sometimes expressed as a *cost* $-r(s, a)$.)
- ▶ **Discount factor $\beta \in (0, 1)$** : Weights future rewards relative to current ones.
- ▶ **Objective**: Choose a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ to maximize expected discounted rewards:

$$\mathbb{E}_{\pi} \left[\sum_{t=0}^T \beta^t r_t(s_t, a_t) + \beta^T R_T(s_T) \right].$$

Recap: Value function and Bellman equation

- ▶ Problem Specification:
 - ▶ Infinite horizon (Finite horizon problems can be solved by backward-induction)
 - ▶ Time-homogeneous: known $P(s' | s, a), r(s' | s, a)$
 - ▶ Didn't specify **State space $\mathcal{S}$, Action space $\mathcal{A}(s)$** to be continuous or discrete
- ▶ Two versions of the Bellman equations:
 - ▶ Policy evaluation for a given π :

$$V^\pi(s) = r(s, \pi(s)) + \beta \sum_{s'} P(s' | s, \pi(s)) V^\pi(s'), \quad \forall s \in \mathcal{S}.$$

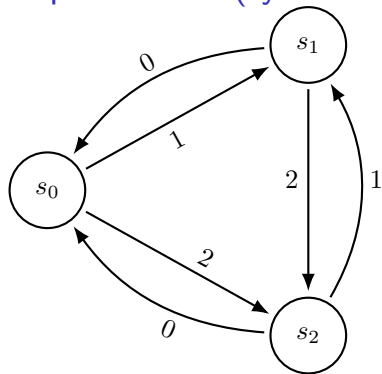
- ▶ Optimality:

$$V^*(s) = \max_{a \in \mathcal{A}(s)} \left\{ r(s, a) + \beta \sum_{s'} P(s' | s, a) V^*(s') \right\}, \quad \forall s \in \mathcal{S}.$$

where $V^* = V^{\pi^*}$ and

$$\pi^*(s) \in \arg \max_{a \in \mathcal{A}(s)} \left\{ r(s, a) + \beta \sum_{s'} P(s' | s, a) V^*(s') \right\}, \quad \forall s \in \mathcal{S}.$$

Example: Jacobi (synchronous) Policy Evaluation


$$\pi =$$

	a_0	a_1	a_2
s_0	0	$\frac{1}{2}$	$\frac{1}{2}$
s_1	$\frac{1}{2}$	0	$\frac{1}{2}$
s_2	$\frac{1}{2}$	$\frac{1}{2}$	0

Initialization:

$$V^0(s_0) = V^0(s_1) = V^0(s_2) = 0$$

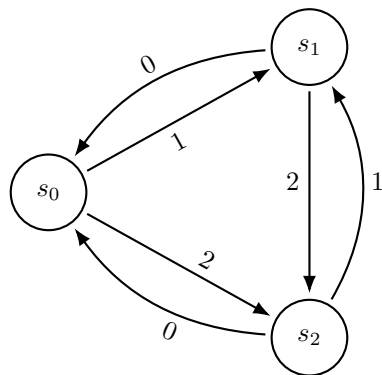
Iteration 1:

$$\begin{aligned} V^1(s_0) &= (T^\pi V^0)(s_0) \\ &= \frac{1}{2}(1 + \beta V^0(s_1)) + \frac{1}{2}(2 + \beta V^0(s_2)) = 1.5 \end{aligned}$$

$$\begin{aligned} V^1(s_1) &= (T^\pi V^0)(s_1) \\ &= \frac{1}{2}(0 + \beta V^0(s_0)) + \frac{1}{2}(2 + \beta V^0(s_2)) = 1 \end{aligned}$$

$$\begin{aligned} V^1(s_2) &= (T^\pi V^0)(s_2) \\ &= \frac{1}{2}(0 + \beta V^0(s_0)) + \frac{1}{2}(1 + \beta V^0(s_1)) = 0.5 \end{aligned}$$

Example: Jacobi (synchronous) Policy Evaluation



Iteration 2:

$$\begin{aligned} V^2(s_0) &= (T^\pi V^1)(s_0) \\ &= \frac{1}{2}(1 + \beta V^1(s_1)) + \frac{1}{2}(2 + \beta V^1(s_2)) = 2.175 \end{aligned}$$

$$\begin{aligned} V^2(s_1) &= (T^\pi V^1)(s_1) \\ &= \frac{1}{2}(0 + \beta V^1(s_0)) + \frac{1}{2}(2 + \beta V^1(s_2)) = 1.9 \end{aligned}$$

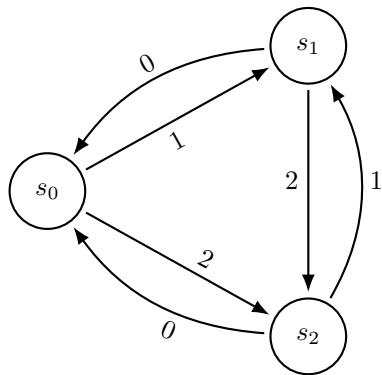
$$\begin{aligned} V^2(s_2) &= (T^\pi V^1)(s_2) \\ &= \frac{1}{2}(0 + \beta V^1(s_0)) + \frac{1}{2}(1 + \beta V^1(s_1)) = 1.625 \end{aligned}$$

...

Iteration 89: converge to tolerance $\|\Delta V\|_\infty < 10^{-4}$

$$V^{89} \approx (10.344, 9.999, 9.654).$$

Example: Gauss–Seidel (in-place / asynchronous) Policy Evaluation



Iteration 0: $V^0(s_0) = V^0(s_1) = V^0(s_2) = 0$.

In-place sweep order $s_0 \rightarrow s_1 \rightarrow s_2$.

Iteration 1:

$$V^1(s_0) = \frac{1}{2}(1 + \beta V^0(s_1)) + \frac{1}{2}(2 + \beta V^0(s_2)) = 1.5$$

$$V^1(s_1) = \frac{1}{2}(0 + \beta V^1(s_0)) + \frac{1}{2}(2 + \beta V^0(s_2)) = 1.675$$

$$V^1(s_2) = \frac{1}{2}(0 + \beta V^1(s_0)) + \frac{1}{2}(1 + \beta V^1(s_1)) = 1.9288$$

Iteration 2:

$$V^2(s_0) = \frac{1}{2}(1 + \beta V^1(s_1)) + \frac{1}{2}(2 + \beta V^1(s_2)) = 3.1217$$

$$V^2(s_1) = \frac{1}{2}(0 + \beta V^2(s_0)) + \frac{1}{2}(2 + \beta V^1(s_2)) = 3.2727$$

$$V^2(s_2) = \frac{1}{2}(0 + \beta V^2(s_0)) + \frac{1}{2}(1 + \beta V^2(s_1)) = 3.3775$$

...

converged in 49 iterations

Policy Evaluation — Linear Algebra (building r and P)

For each state s ,

$$r_s = \sum_a \pi(a|s) R(s, a), \quad P_{s,s'} = \sum_a \pi(a|s) \mathbf{1}\{a \text{ leads to } s'\}.$$

From the diagram (outgoing rewards and next states):

$$\begin{aligned} s_0 : 1 \text{ to } s_1, 2 \text{ to } s_2 &\Rightarrow r_0 = \frac{1+2}{2} = 1.5, & P_{0,1} = P_{0,2} = \frac{1}{2}, & P_{0,0} = 0, \\ s_1 : 0 \text{ to } s_0, 2 \text{ to } s_2 &\Rightarrow r_1 = \frac{0+2}{2} = 1, & P_{1,0} = P_{1,2} = \frac{1}{2}, & P_{1,1} = 0, \\ s_2 : 0 \text{ to } s_0, 1 \text{ to } s_1 &\Rightarrow r_2 = \frac{0+1}{2} = 0.5, & P_{2,0} = P_{2,1} = \frac{1}{2}, & P_{2,2} = 0. \end{aligned}$$

Therefore

$$r = \begin{bmatrix} 1.5 \\ 1 \\ 0.5 \end{bmatrix}, \quad P = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}.$$

Policy Evaluation — linear algebra (solve and spectral radius)

Bellman (policy evaluation):

$$V^* = r + \beta P V^* \iff (I - \beta P) V^* = r.$$

With $\beta = 0.9$,

$$A \equiv I - \beta P = \begin{bmatrix} 1 & -0.45 & -0.45 \\ -0.45 & 1 & -0.45 \\ -0.45 & -0.45 & 1 \end{bmatrix}, \quad V^* = A^{-1}r = \begin{bmatrix} \frac{300}{29} \\ 10 \\ \frac{280}{29} \end{bmatrix} \approx \begin{bmatrix} 10.3448 \\ 10.0000 \\ 9.6552 \end{bmatrix}.$$

- ▶ If $\rho(\beta P) < 1$, then $T(V) = r + \beta P V$ is a contraction: $(I - \beta P)$ is invertible, $(I - \beta P)^{-1} = \sum_{k \geq 0} (\beta P)^k$, and the Bellman solution is unique; value iteration converges.
- ▶ Eigenvalues scale: $\rho(\beta P) = |\beta| \rho(P)$.
- ▶ P is row-stochastic $\Rightarrow \rho(P) = 1$ (here $P = \frac{1}{2}(J - I)$ with eigenvalues $1, -\frac{1}{2}, -\frac{1}{2}$).
- ▶ Hence $\rho(\beta P) = \beta < 1$.

Gauss–Seidel Policy Evaluation in linear algebra form (advanced)

Start from the linear system for policy evaluation:

$$A \mathbf{V} = \mathbf{b}, \quad A := I - \beta P, \quad \mathbf{b} := \mathbf{r}.$$

Split A into diagonal, strictly lower, and strictly upper parts:

$$A = D + L + U, \quad D := \text{diag}(A), \quad L := \text{tril}(A, -1), \quad U := \text{triu}(A, 1).$$

Jacobi (synchronous) iteration (for comparison):

$$\mathbf{V}^{k+1} = D^{-1}(\mathbf{b} - (L + U) \mathbf{V}^k) \quad (\text{uses only old values on the RHS; parallel-friendly}).$$

Gauss–Seidel (in-place) iteration:

$$(D + L) \mathbf{V}^{k+1} = U \mathbf{V}^k + \mathbf{b} \quad \Longleftrightarrow \quad \mathbf{V}^{k+1} = (D + L)^{-1}(\mathbf{b} - U \mathbf{V}^k).$$

- ▶ GS solves a lower-triangular system each sweep (forward substitution), so rows use newest components as soon as they're available.
- ▶ Typically fewer iterations than Jacobi; depends on ordering and sparsity.

Direct policy evaluation summary

Idea: For a fixed policy π , the Bellman equation is linear:

$$V = r + \beta PV \iff AV = b, A := I - \beta P, b := r.$$

Intuition: This “sums the whole infinite discounted future” in one go.

When to use:

- ▶ Small/medium state spaces; A not too dense.
- ▶ You will evaluate the same P, β for many different reward vectors r
- ▶ You want tight, predictable control of the residual $\|AV - b\|$.

Trade-offs:

- ▶ Setup cost and memory for factorization; sparse A can “fill in”.
- ▶ If the policy changes (so P changes), you typically must refactor.
- ▶ Handles evaluation only; the $\max$ in control (optimality) is not a single linear solve.

Iterative policy evaluation summary

Idea: Start from a guess $V^{(0)}$ and repeatedly apply the policy-Bellman map

Jacobi: $V^{(k+1)} = r + \beta P V^{(k)}$ (update all states from the old vector);

Intuition: “Plug the current guess back into the right-hand side” and repeat.

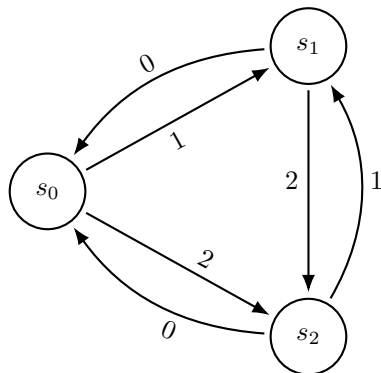
When to use:

- ▶ Large state spaces; P is sparse; you want matrix-free updates and low memory.
- ▶ Policies/rewards change often (easy warm starts from the last V).
- ▶ You plan to move on to controlw: the same style extends to optimality (max) updates.

Trade-offs:

- ▶ Need a stopping rule ($\|V^{(k+1)} - V^{(k)}\|_\infty$ or residual); more sweeps as $\beta \rightarrow 1$.

Example: Jacobi (synchronous) Value Iteration — Optimal Policy



Initialization:

$$V^0(s_0) = V^0(s_1) = V^0(s_2) = 0.$$

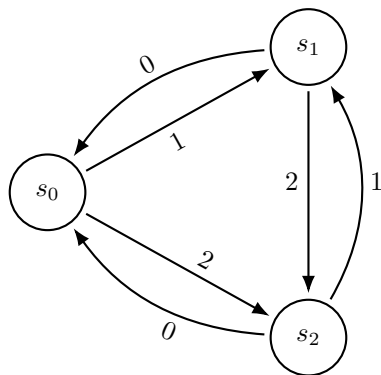
Iteration 1:

$$\begin{aligned} V^1(s_0) &= \max\{1 + 0.9 V^0(s_1), 2 + 0.9 V^0(s_2)\} \\ &= \max\{1, 2\} = 2 \end{aligned}$$

$$\begin{aligned} V^1(s_1) &= \max\{0 + 0.9 V^0(s_0), 2 + 0.9 V^0(s_2)\} \\ &= \max\{0, 2\} = 2 \end{aligned}$$

$$\begin{aligned} V^1(s_2) &= \max\{0 + 0.9 V^0(s_0), 1 + 0.9 V^0(s_1)\} \\ &= \max\{0, 1\} = 1. \end{aligned}$$

Example: Jacobi (synchronous) Value Iteration — Optimal Policy



Iteration 2:

$$\begin{aligned} V^2(s_0) &= \max\{1 + \beta V^1(s_1), 2 + \beta V^1(s_2)\} \\ &= \max\{1 + 0.9 \cdot 2, 2 + 0.9 \cdot 1\} = 2.9 \end{aligned}$$

$$\begin{aligned} V^2(s_1) &= \max\{0 + \beta V^1(s_0), 2 + \beta V^1(s_2)\} \\ &= \max\{0.9 \cdot 2, 2 + 0.9 \cdot 1\} = 2.9 \end{aligned}$$

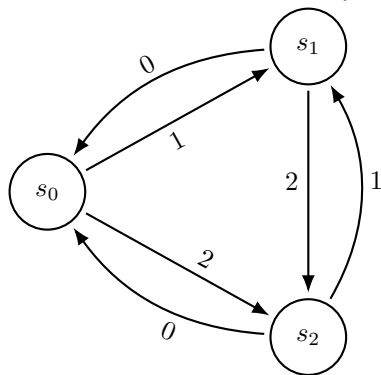
$$\begin{aligned} V^2(s_2) &= \max\{0 + \beta V^1(s_0), 1 + \beta V^1(s_1)\} \\ &= \max\{0.9 \cdot 2, 1 + 0.9 \cdot 2\} = 2.8 \end{aligned}$$

...

Iteration 95: converge to tolerance $\|\Delta V\|_\infty < 10^{-4}$

$$V^{95} \approx (15.263, 15.263, 14.737).$$

Example: Gauss–Seidel (in-place) Value Iteration — Optimal Policy



Iteration 0: $V^0(s_0) = V^0(s_1) = V^0(s_2) = 0$.

Iteration 1:

$$V^1(s_0) = \max\{1 + \beta V^0(s_1), 2 + \beta V^0(s_2)\} = 2$$

$$V^1(s_1) = \max\{0 + \beta V^1(s_0), 2 + \beta V^0(s_2)\} = 2$$

$$V^1(s_2) = \max\{0 + \beta V^1(s_0), 1 + \beta V^1(s_1)\} = 2.8$$

Iteration 2:

$$V^2(s_0) = \max\{1 + \beta V^1(s_1), 2 + \beta V^1(s_2)\} = 4.52$$

$$V^2(s_1) = \max\{0 + \beta V^2(s_0), 2 + \beta V^1(s_2)\} = 4.52$$

$$V^2(s_2) = \max\{0 + \beta V^2(s_0), 1 + \beta V^2(s_1)\} = 5.068$$

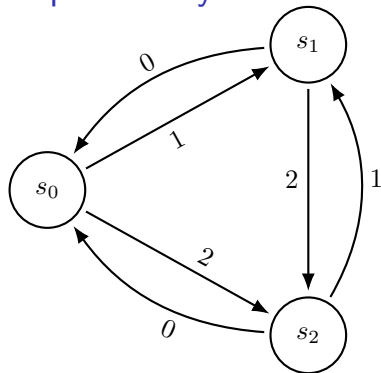
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Iteration 51: converge to tolerance $\|\Delta V\|_\infty < 10^{-4}$

$$V^{51} \approx (15.263, 15.263, 14.737).$$

Example: Policy Function Iteration (PFI)

Iteration 0: initialize policy


$$\pi^0 =$$

	a_0	a_1	a_2
s_0	0	$\frac{1}{2}$	$\frac{1}{2}$
s_1	$\frac{1}{2}$	0	$\frac{1}{2}$
s_2	$\frac{1}{2}$	$\frac{1}{2}$	0

Iteration 1:

Step 1: Policy evaluation: Converged in 49 iterations to

$$V^{\pi^0} = (10.3448, 10.0000, 9.6552)$$

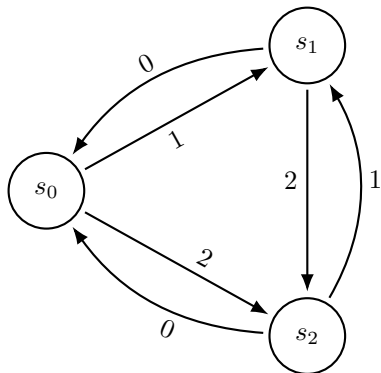
Step 2: Policy improvement (greedy w.r.t. V^{π^0})

$$\pi^1(s_0) = \arg \max\{1 + \beta V_1^{\pi^0}, 2 + \beta V_2^{\pi^0}\} = a_2$$

$$\pi^1(s_1) = \arg \max\{0 + \beta V_0^{\pi^0}, 2 + \beta V_2^{\pi^0}\} = a_2$$

$$\pi^1(s_2) = \arg \max\{0 + \beta V_0^{\pi^0}, 1 + \beta V_1^{\pi^0}\} = a_1$$

Example: Policy Function Iteration (PFI)



Iteration 2: policy $\pi^1 = (a_2, a_2, a_1)$

Step 1: Policy evaluation: Start from V^{π^0} as initial guess. Converged in 46 iterations to

$$V^{\pi^1} = (15.2632, 15.2632, 14.7368).$$

Step 2: Policy improvement (greedy w.r.t. V^{π^1})

$$\pi^2(s_0) = \arg \max\{1 + \beta V_1^{\pi^1}, 2 + \beta V_2^{\pi^1}\} = a_2$$

$$\pi^2(s_1) = \arg \max\{0 + \beta V_0^{\pi^1}, 2 + \beta V_2^{\pi^1}\} = a_2$$

$$\pi^2(s_2) = \arg \max\{0 + \beta V_0^{\pi^1}, 1 + \beta V_1^{\pi^1}\} = a_1$$

$\Rightarrow \boxed{\pi^2 = \pi^1}$ (policy converged) $\Rightarrow$ Stop.

Value Function Iteration (time-homogeneous, infinite horizon)

- ▶ Step 0 (inputs): state space $\mathcal{S}$; actions $\mathcal{A}(s)$; reward $r(s, a)$; transition $P(\cdot \mid s, a)$; discount $\beta \in (0, 1)$; tolerance $\varepsilon > 0$.

- ▶ Step 1 (Bellman operator): for any $V : \mathcal{S} \rightarrow \mathbb{R}$,

$$(TV)(s) = \max_{a \in \mathcal{A}(s)} \left\{ r(s, a) + \beta \mathbb{E}[V(S') \mid s, a] \right\}.$$

- ▶ Step 2 (initialize value): choose $V^{(0)}$ (e.g., $V^{(0)} \equiv 0$); set $k \leftarrow 0$.
- ▶ Step 3 (value update): for each state s ,

$$Q_{V^{(k)}}(s, a) = r(s, a) + \beta \mathbb{E}[V^{(k)}(S') \mid s, a] \quad \text{for all } a \in \mathcal{A}(s),$$

$$V^{(k+1)}(s) \leftarrow \max_{a \in \mathcal{A}(s)} Q_{V^{(k)}}(s, a).$$

- ▶ Step 4 (convergence check): compute $\Delta_{k+1} = \|V^{(k+1)} - V^{(k)}\|_{\infty}$. If $\Delta_{k+1} \leq \varepsilon$, go to Step 5; else set $k \leftarrow k + 1$ and return to Step 3.
- ▶ Step 5 (policy extraction): greedy w.r.t. $Q_{V^{(k+1)}}$,

$$\pi^*(s) \in \arg \max_{a \in \mathcal{A}(s)} \left\{ r(s, a) + \beta \mathbb{E}[V^{(k+1)}(S') \mid s, a] \right\}.$$

Policy Function Iteration (time-homogeneous, infinite horizon)

- ▶ Step 0 (inputs): state space $\mathcal{S}$; actions $\mathcal{A}(s)$; reward $r(s, a)$; transition $P(\cdot \mid s, a)$; discount $\beta \in (0, 1)$; tolerances $\varepsilon_{\text{eval}}, \varepsilon_{\text{imp}} > 0$.

- ▶ Step 1 (policy Bellman operator): for a stationary policy π and any $V : \mathcal{S} \rightarrow \mathbb{R}$,

$$(T^\pi V)(s) = r(s, \pi(s)) + \beta \mathbb{E}[V(S') \mid s, \pi(s)].$$

- ▶ Step 2 (initialize policy): choose any feasible stationary policy $\pi^{(0)}$; set $k \leftarrow 0$.

- ▶ Step 3 (policy evaluation): compute $V^{\pi^{(k)}}$ as the fixed point of $T^{\pi^{(k)}}$,

$$V^{\pi^{(k)}} = T^{\pi^{(k)}} V^{\pi^{(k)}},$$

either exactly (linear solve in finite MDPs) or iteratively until $\|T^{\pi^{(k)}} V - V\|_\infty \leq \varepsilon_{\text{eval}}$.

- ▶ Step 4 (policy improvement): for each $s \in \mathcal{S}$,

$$\pi^{(k+1)}(s) \in \arg \max_{a \in \mathcal{A}(s)} \left\{ r(s, a) + \beta \mathbb{E}[V^{\pi^{(k)}}(S') \mid s, a] \right\}.$$

- ▶ Step 5 (convergence check): if $\pi^{(k+1)} = \pi^{(k)}$ (or the improvement gain $\leq \varepsilon_{\text{imp}}$), go to Step 6; else set $k \leftarrow k + 1$ and return to Step 3.

- ▶ Step 6 (outputs): optimal policy $\pi^* = \pi^{(k)}$ and value $V^* = V^{\pi^{(k)}}$.

Decomposing value iteration into evaluation + improvement

Value iteration (one step):

$$V^{(k+1)} = (TV^{(k)})(s) = \max_{a \in \mathcal{A}(s)} \{r(s, a) + \beta \mathbb{E}[V^{(k)}(S') \mid s, a]\}.$$

Equivalent two-step decomposition:

1. improvement (greedy w.r.t. $V^{(k)}$):

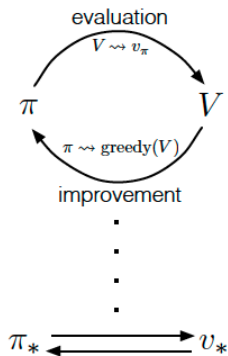
$$\pi^{(k)}(s) \in \arg \max_{a \in \mathcal{A}(s)} \{r(s, a) + \beta \mathbb{E}[V^{(k)}(S') \mid s, a]\};$$

2. one-step policy evaluation:

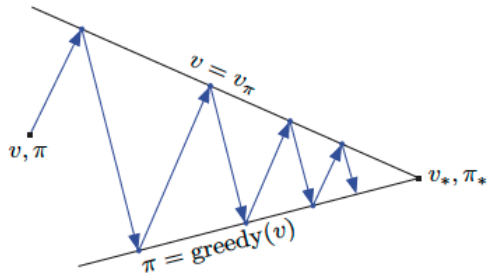
$$V^{(k+1)}(s) = (T^{\pi^{(k)}} V^{(k)})(s) = r(s, \pi^{(k)}(s)) + \beta \mathbb{E}[V^{(k)}(S') \mid s, \pi^{(k)}(s)].$$

Notes: Full evaluation to $V^{\pi^{(k)}}$ yields classical policy iteration; a single backup yields value iteration; in-between gives modified policy iteration. All are instances of GPI.

Generalized Policy Iteration: two views



(a) View 1



(b) View 2

Figure 1: Generalized Policy Iteration (two views), adapted from [Sutton and Barto \(2018\)](#), Chapter 4.

Howard Policy Iteration (PFI/VFI hybrid)

- ▶ Step 0 (inputs): as in PFI, plus an integer $m \geq 1$ (number of evaluation sweeps per improvement) and tolerance $\varepsilon > 0$.
- ▶ Step 1 (initialize): choose $\pi^{(0)}$ and $V^{(0)}$; set $k \leftarrow 0$.
- ▶ Step 2 (partial policy evaluation): set $U^{(0)} \leftarrow V^{(k)}$; for $i = 0, 1, \dots, m-1$,

$$U^{(i+1)} \leftarrow T^{\pi^{(k)}} U^{(i)}.$$

Set $V^{(k+1)} \leftarrow U^{(m)}$.

- ▶ Step 3 (policy improvement): for each s ,

$$\pi^{(k+1)}(s) \in \arg \max_{a \in \mathcal{A}(s)} \left\{ r(s, a) + \beta \mathbb{E} \left[V^{(k+1)}(S') \mid s, a \right] \right\}.$$

- ▶ Step 4 (convergence check): if $\|V^{(k+1)} - V^{(k)}\|_{\infty} \leq \varepsilon$ and $\pi^{(k+1)} = \pi^{(k)}$, stop; else set $k \leftarrow k + 1$ and return to Step 2.
- ▶ Notes: $m = 1$ recovers value iteration (VFI-style single backup before improvement); $m = \infty$ recovers policy function iteration with full evaluation (PFI/Howard).

Approximation in Dynamic Programming

approximate max

Discretization /
Selective Minimization /
Gradient

at s_t

$$\max_{a_t \in \mathcal{A}_t(s_t)} \mathbb{E} \left[r_t(s_t, a_t, W_t) + \beta \tilde{V}_{t+1}(S_{t+1}) \mid s_t, a_t \right]$$

approximate $\mathbb{E}$

Monte Carlo / Quasi-MC
/ Quadrature / Sparse
sampling

approximate $\tilde{V}_{t+1}$

Parametric approximation /
Neural networks /
Problem approximation /
Rollout

References & Further Reading

- ▶ Dimitri P. Bertsekas, **Reinforcement learning and optimal control (2025 Spring course at ASU)**, Lecture 3:
<https://web.mit.edu/dimitrib/www/RLbook.html>
- ▶ Sutton and Barto, **Reinforcement Learning: An Introduction**, Chapter 4 and 5,
<https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf>

Sutton, Richard S. and Andrew G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.